

Source Code Management Lab

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Section: CSE 4th Sem

Course: Source Code Management

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GIT INSTALLATION

Step 1: Downloading Git

1. Open your web browser and navigate to the official Git website: <https://git-scm.com>.
2. On the homepage, you will see a **"Download"** button that automatically detects your OS. Click on the "Download" button to download the appropriate installer for your operating system (Windows, macOS, or Linux).
3. Alternatively, you can manually select your OS from the website to download a specific version.



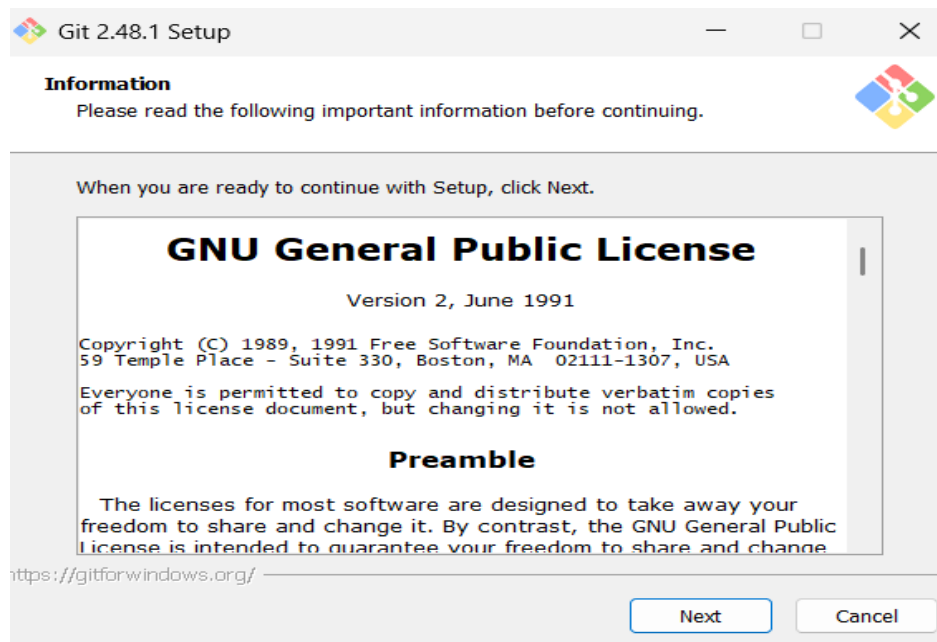
Step 2: Running the Git Installer

Locate the downloaded Git.exe file and double-click to run it.

Name	Date modified	Type	Size
HP Downloads	22-07-2024 17:20	File folder	
Support-LogMeInRescue	27-08-2022 13:18	Application	2,139 KB
sp141572	29-08-2022 13:35	Application	63,879 KB
matlab_R2022b_win64	07-11-2022 09:40	Application	2,33,022 KB
sp153036	11-07-2024 13:32	Application	24,809 KB
python-3.12.5-amd64	02-09-2024 18:30	Application	25,888 KB
Firefox Installer	11-10-2024 23:27	Application	365 KB
VSCodeUserSetup-x64-1.96.4	19-01-2025 15:51	Application	1,02,501 KB
avast_free_antivirus_setup_online	19-01-2025 16:02	Application	244 KB
python-3.13.1-amd64	27-01-2025 15:03	Application	28,016 KB
CLion-2024.3.3	21-02-2025 22:26	Application	13,61,215 ...
pycharm-professional-2024.3.3	21-02-2025 22:33	Application	8,56,782 KB
idealIU-2024.3.3	21-02-2025 22:34	Application	11,85,621 ...
Git-2.48.1-64-bit	27-02-2025 19:02	Application	67,915 KB

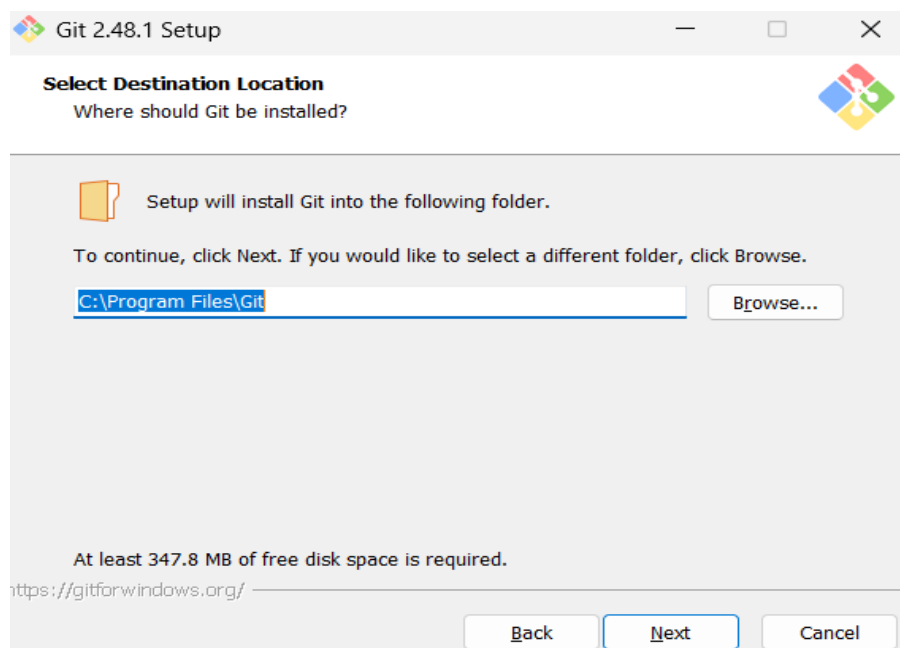
Step 3: License (Terms and Conditions)

Read the GNU General Public License's terms and conditions and click on **Next**.



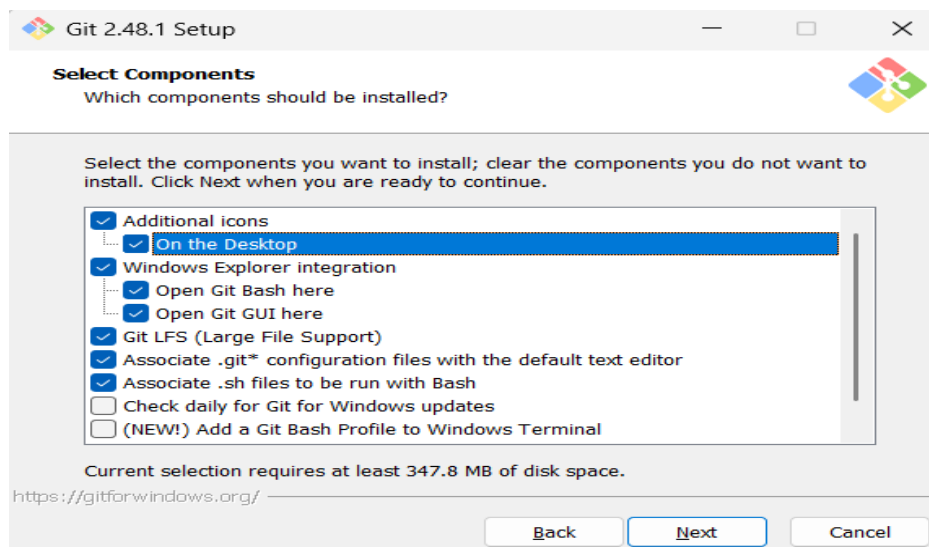
Step 4: Choose Installation Location

Choose the installation location (default is **C:\Program Files\Git**) and click **Next**.



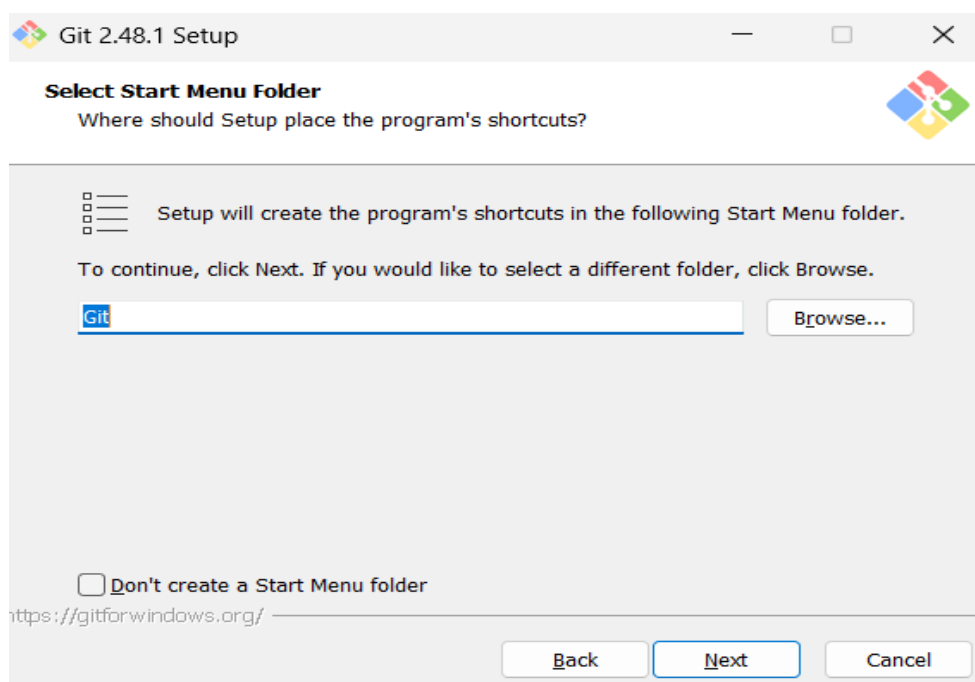
Step 5: Select the Components

Select the components you want (default options are fine) and click **Next**.



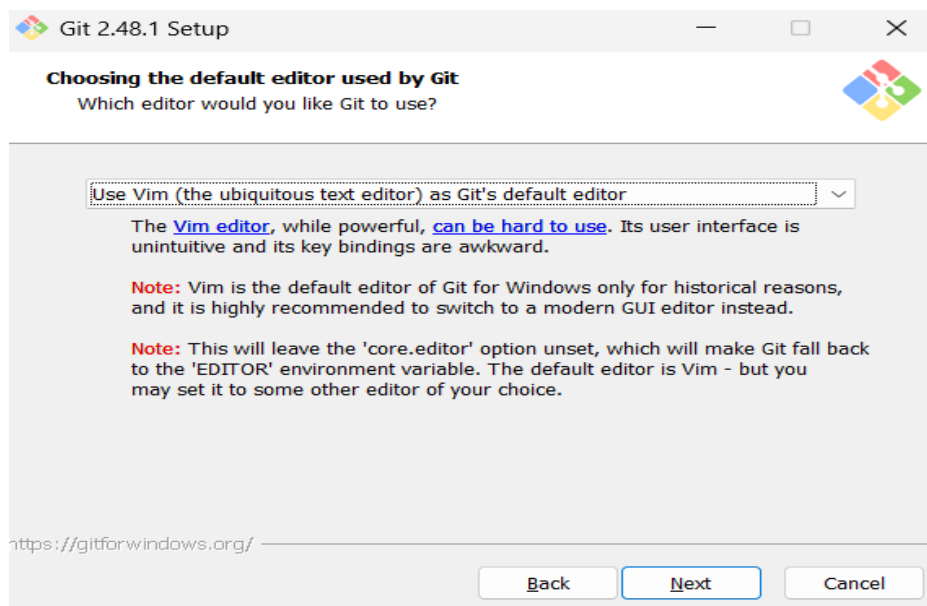
Step 6: Select Start Menu Folder

Choose the Start Menu folder where Git shortcuts will be placed. By default, the folder is named **"Git"**. Keep the default name and click **Next** to Proceed.



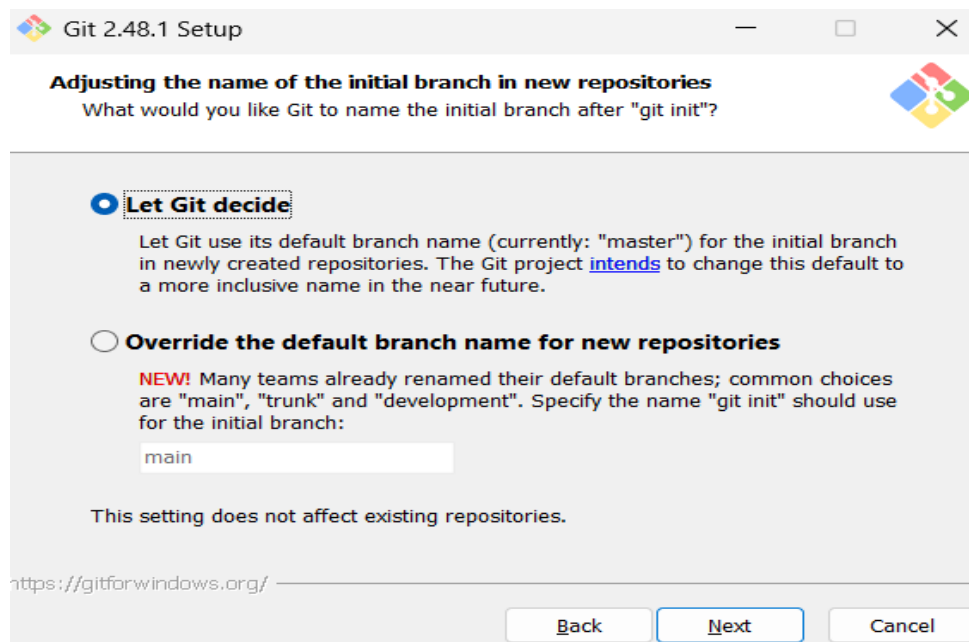
Step 7: Choose the Text Editor

Choose a default text editor (select **Vim**) and Click **Next**.



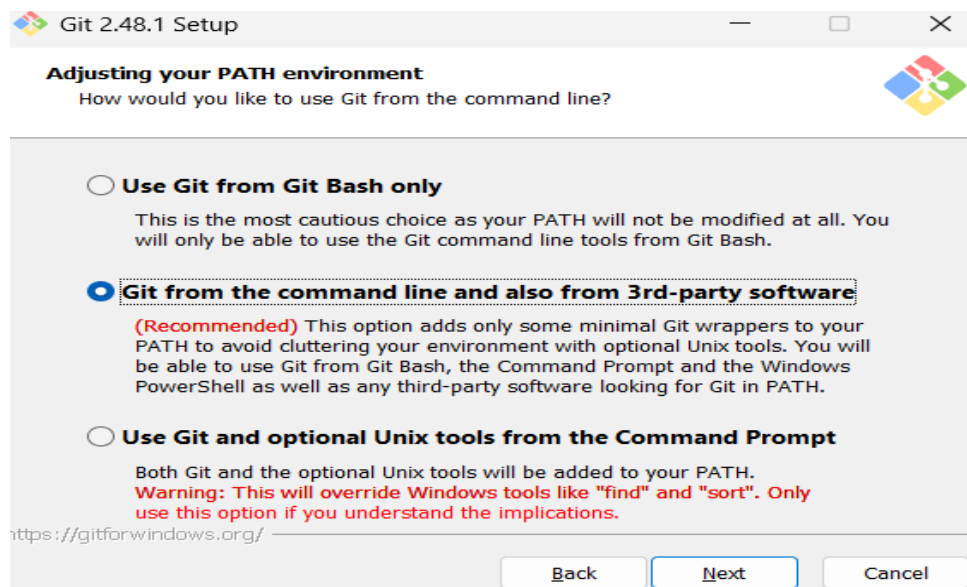
Step 8: Adjusting Initial Branch Name

Choose the default name for the first branch when initializing a new Git repository. Go with **'Let Git Decide'** option setting the branch as **Master** branch and proceed with **Next**.



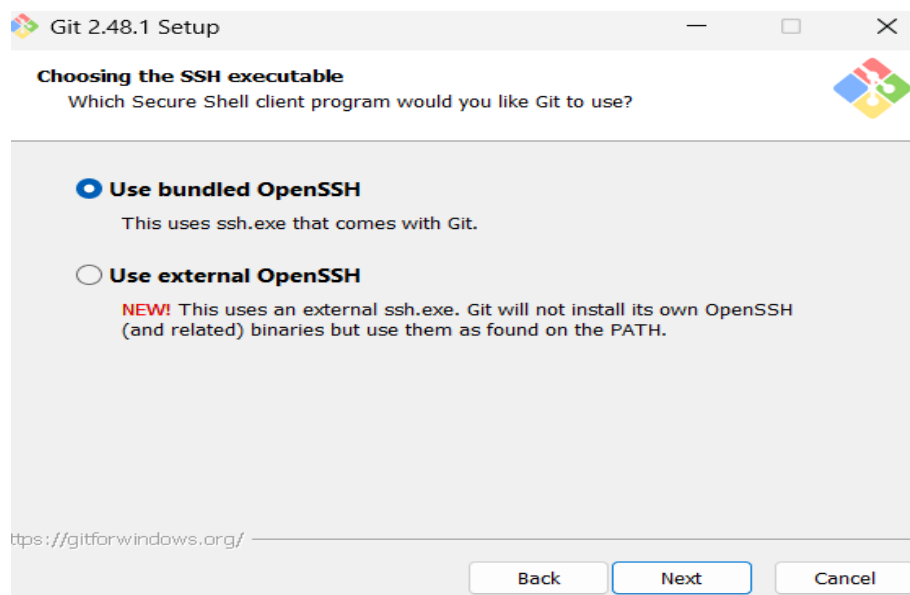
Step 9: Adjusting PATH Environment

Select **Git** from the command line and also from third-party software (recommended). Click **Next**.



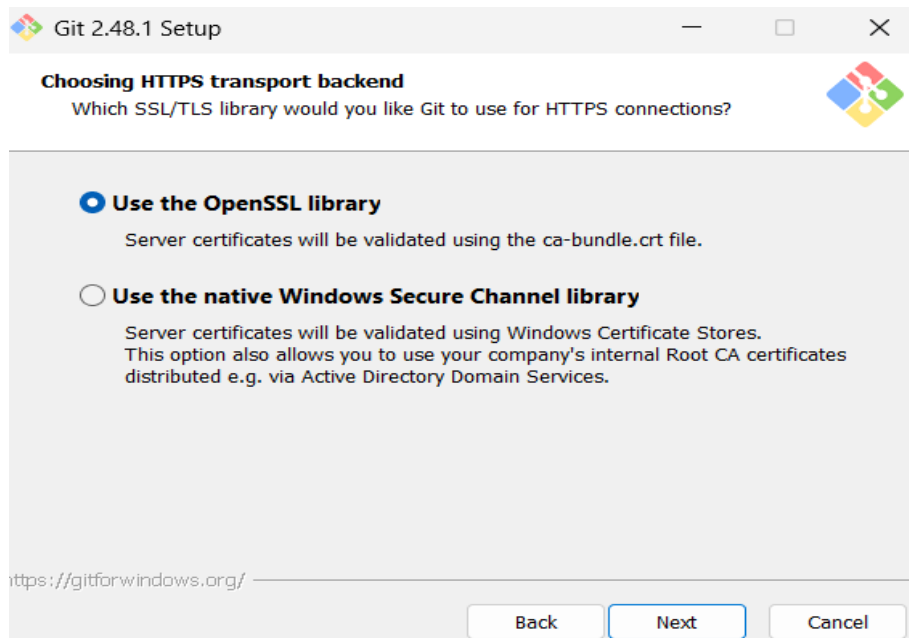
Step 10: Choosing the SSH Executable

Select "**Use bundled OpenSSH**" for better compatibility and Click on **Next**.



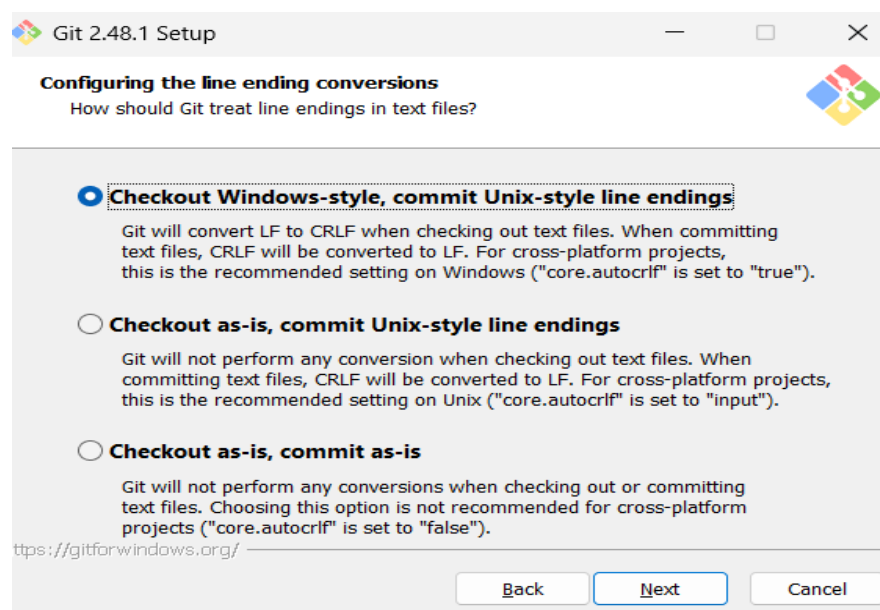
Step 11: Choosing the HTTP Transport Background

Choose Use the **OpenSSL** library (default) and Click **Next**.



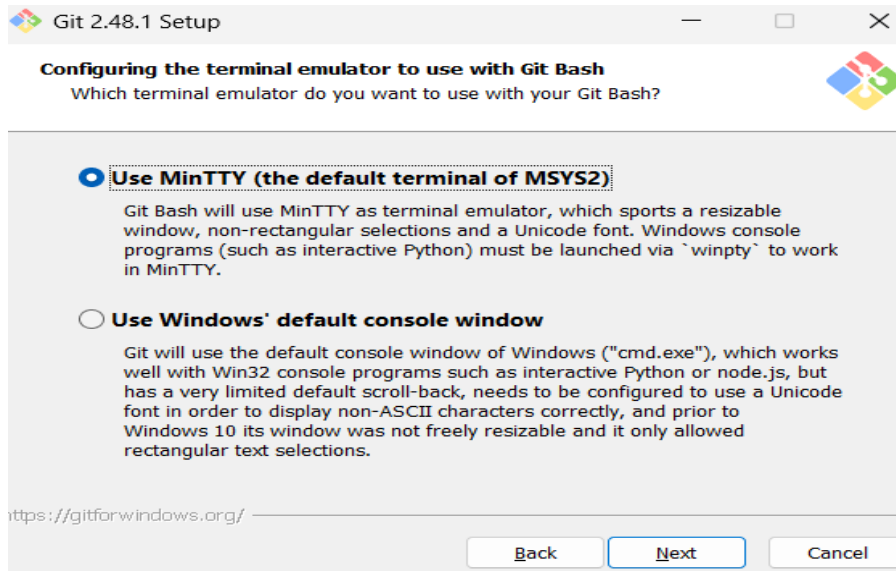
Step 12: Configuring Line Ending Configs

Select **Checkout Windows-style, commit Unix-style line endings** (recommended) and Click **Next**.



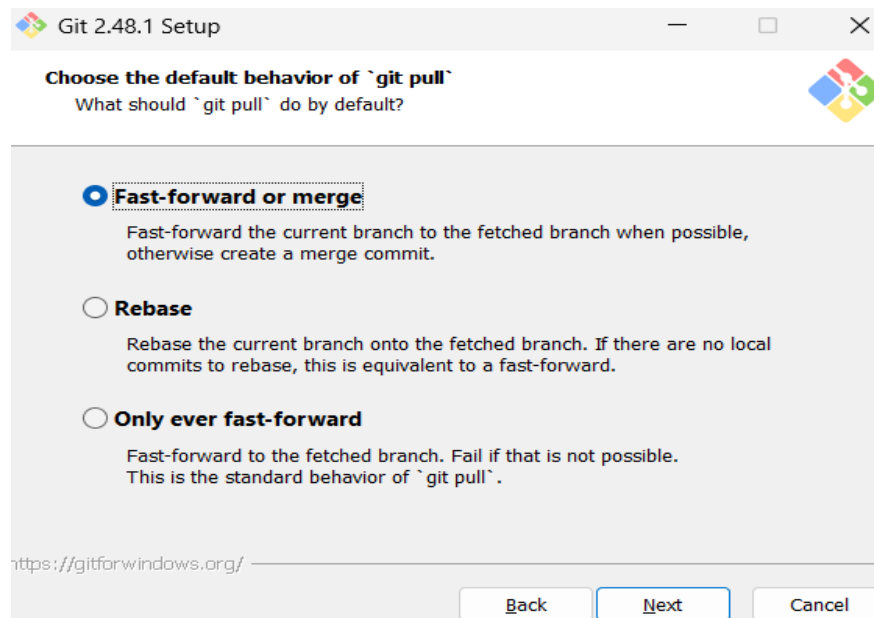
Step 13: Configuring the Terminal Emulator

Select **Use MinTTY (default terminal for MSYS2)** and Click **Next**.



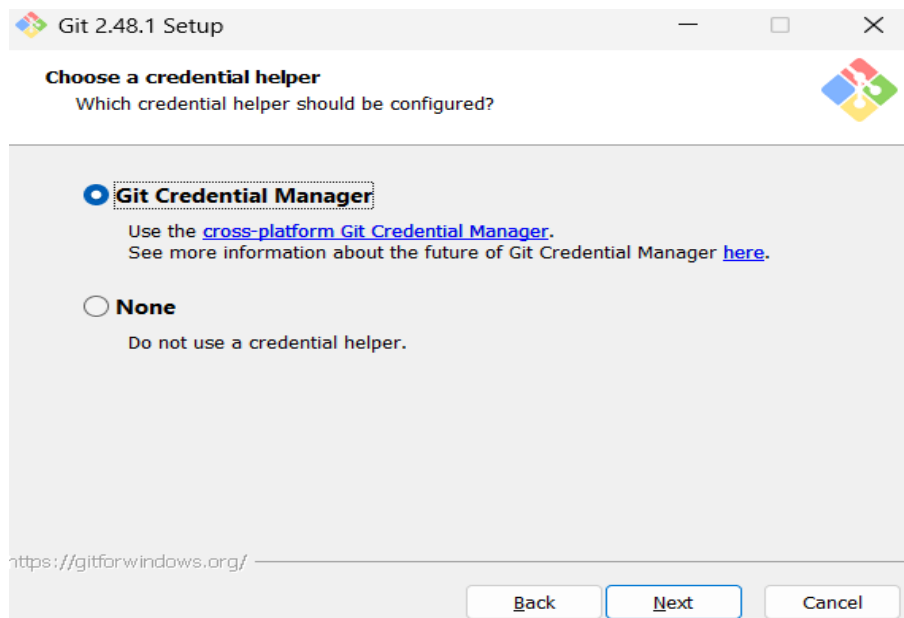
Step 14: Choosing the Default Behaviour

Select **Fast-forward or Merge** (recommended) option and click **Next**.



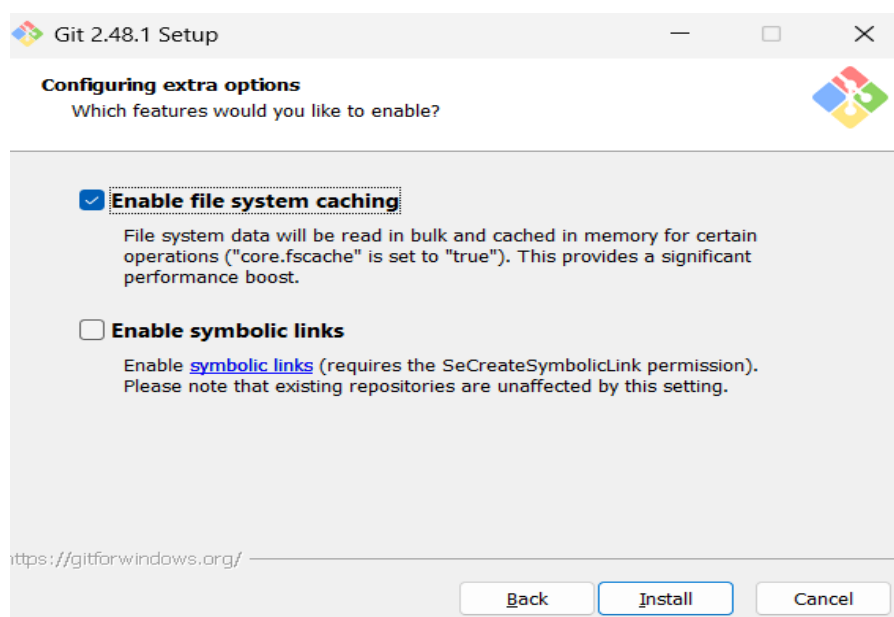
Step 15: Choosing a Credential Helper

Select **Git Credential Manager** (recommended) and Click **Next**.



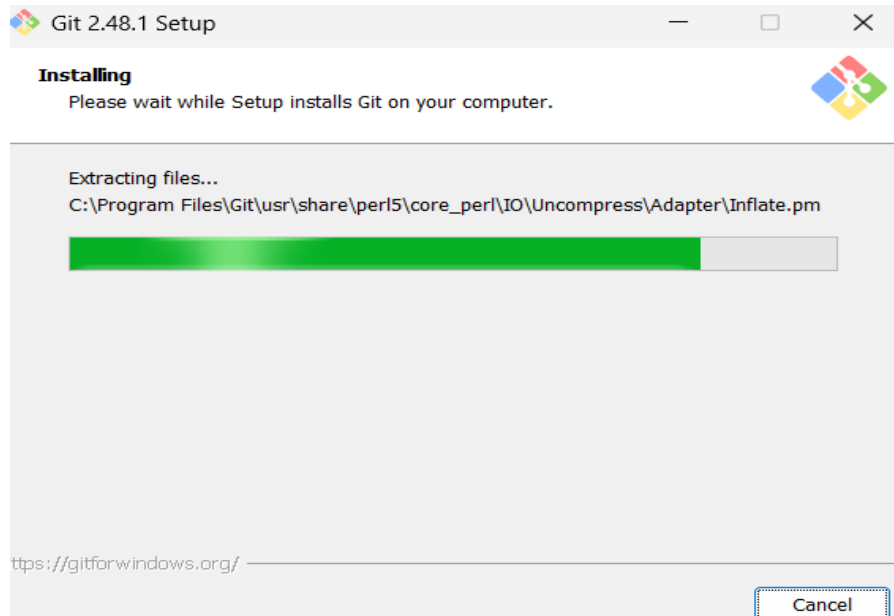
Step 16: Configuring Extra Options

Select **Enable file system caching** (recommended) and Click on **Install**.



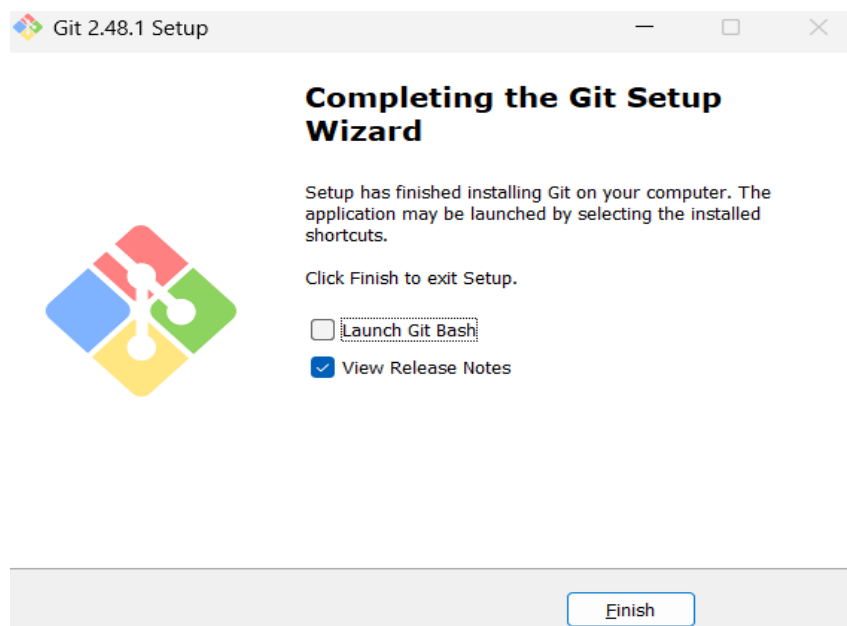
Step 17: Installation Overview

A progress bar (**green bar**) will appear, indicating that Git is being installed. Wait for the installation to complete. This may take a few minutes.



Step 18: Completing the Git Set - Up Wizard

Once the installation is complete, "Completing the Git Setup Wizard" screen appears. Check the 'Launch Git bash' option and Click on **Finish**.



COMMANDS IN GIT

1. pwd

- *Purpose:* Displays the Present Working Directory.
- *Usage:* Use this to find the directory you're currently working in.

```
Admin@DESKTOP-OD400I1 MINGW64 ~  
$ pwd  
/c/Users/Admin
```

2. **cd**

- *Purpose:* Change directory.
- *Usage:* Navigate to a different directory

```
Admin@DESKTOP-OD400I1 MINGW64 ~  
$ cd ..
```

3. **ls**

- *Purpose:* Lists all folders in the current directory.

```
Admin@DESKTOP-OD400I1 MINGW64 /c/Users  
$ ls  
Admin/ 'All Users'@ Default/ 'Default User'@ Public/ desktop.ini
```

4. **vi**

- *Purpose:* Open the Vim/Vi editor to create or edit files.
- *Usage:* vi filename. Press I to insert, esc to exit insert mode, and :wq to save and quit.

5. **cat**

- *Purpose:* View file contents.
- *Usage:* cat filename.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test  
$ vi test.txt  
  
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test  
$ cat test.txt  
Just A Trial
```

6. **ls -l**

- *Purpose:* Lists files in long format with additional details (permissions, owners, size).

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test
$ ls -l
total 1
-rw-r--r-- 1 Admin 197121 13 Jun  5 23:53 test.txt
```

7. ls -ah

- *Purpose:* Lists all files, including hidden ones.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test
$ ls -ah
./  ../  .swp  test.txt
```

8. history

- *Purpose:* Displays the history of previously run commands.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test
$ history
 1 git config --global.username "Peeyush"
 2 git config --global user.name "Peeyush"
 3 git config --global user.email "peeyushrampal31@gmail.com"
 4 mkdir GitDemo
 5 cd GitDemo
 6 git init
 7 git status
 8 git status
 9 git add sum.py
10 git commit -m "Updated"
11 git branch dev
12 git checkout dev
13 git status
14 git add sum.py
15 git merge dev
16 git merge master
17 git status
18 git checkout master
19 git merge dev
20 git checkout dev
21 git status
22 git commit -m "New update"
23 git merge master
24 git merge dev
--
```

9. mkdir

- *Purpose:* Create a new folder.
- *Usage:* mkdir foldername.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop
$ mkdir test
```

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop
$ cd test
```

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test
$
```

10. git init

- *Purpose:* Initializes an empty Git repository in the current directory.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test
$ git init
Initialized empty Git repository in C:/Users/Admin/Desktop/test/.git/
```

11. git status

- *Purpose:* Displays the current status of the working directory and staging area, showing:
- Modified files not yet staged.
- Staged files ready for commit.
- Untracked files.
- Current branch and whether it's up-to-date with the remote branch.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .swp
        test.txt

nothing added to commit but untracked files present (use "git add" to track)
```

12. git add

- *Purpose:* Stage a file to prepare it for commit.
- *Usage:* git add filename.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git add .
warning: in the working copy of 'test.txt', LF will be replaced by CRLF the next time Git touches it
```

13. git commit

- *Purpose:* Commit changes to the repository with a message.
- *Usage:* git commit -m "Message".

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git commit -m "Initial Commit"
[master (root-commit) 57ab790] Initial Commit
2 files changed, 1 insertion(+)
create mode 100644 .swp
create mode 100644 test.txt
```

13. git config

- *Purpose:* Configure Git with username and email.
- *Usage:*
 - `git config --global user.mail "email@example.com"`
 - `git config --global user.name "Your Name"`

14. git log

- *Purpose:* View commit history.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git log
commit 57ab790a925c97cd178a229b16f89390b791d6bc (HEAD -> master)
Author: Peeyush Rampal <peeyushrampal31@gmail.com>
Date: Thu Jun 5 23:58:10 2025 +0530

    Initial Commit
```

15. git diff

- *Purpose:* shows the changes made in the working directory since the last commit.

16. git --oneline

Purpose: View a concise commit history

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git log --oneline
57ab790 (HEAD -> master) Initial Commit
```

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git diff bae79c8 57ab790
```

```
@@ -1 +1 @@
-Just A Trial and works Perfectly
+Just A Trial
```

17. git branch <branch name>

- *Purpose:* Creates a new branch.

18. git branch

- *Purpose:* Lists all branches in the repository.

19. **git checkout <branch name>**

- *Purpose:* Switch to the master branch.
- *Usage:*

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git branch test
```

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git checkout test
Switched to branch 'test'
```

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (test)
$ git branch
  master
* test
```

20. **git remote add <url>**

- *Purpose:* Connects your local repository to a remote repository.

21. **git remote**

- *Purpose:* Lists the short names of all configured remote repositories.

22. **git remote -v**

- *Purpose:* Displays the URLs of all remote repositories linked to your local repository, showing both **fetch** and **push** directions.

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git remote add origin https://github.com/Peeyush2005/A-Fresh-Repo.git
```

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git remote
origin
```

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git remote -v
origin https://github.com/Peeyush2005/A-Fresh-Repo.git (fetch)
origin https://github.com/Peeyush2005/A-Fresh-Repo.git (push)
```

23. **git push**

- *Purpose:* Pushes the local main branch to the remote repository named origin


```

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git remote add origin https://github.com/Peeyush2005/A-Fresh-Repo.git

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git remote
origin

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ git remote -v
origin https://github.com/Peeyush2005/A-Fresh-Repo.git (fetch)
origin https://github.com/Peeyush2005/A-Fresh-Repo.git (push)

```

GIT MERGE

- **Production Code** is the finalized, deployable version of software used in a live environment.
- **Branches** in Git allow multiple versions of code to exist simultaneously, starting from the project's inception.
- **Merging** happens when different branches need to be combined. Git does not automatically decide what to keep—it requires manual resolution.
- **Conflict Resolution:** If conflicts arise, Git will halt the merge. A merge tool helps resolve differences.

```

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ vi Index.html

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ vi Style.css

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ git add .
warning: in the working copy of 'Index.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Style.css', LF will be replaced by CRLF the next time Git touches it

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ git commit -m "Master 1 commit"
[master (root-commit) d5572ca] Master 1 commit
2 files changed, 13 insertions(+)
create mode 100644 Index.html
create mode 100644 Style.css

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ git checkout -b test
Switched to a new branch 'test'

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (test)
$ vi Index.html

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (test)
$ vi Style.css

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (test)
$ git add .
warning: in the working copy of 'Index.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Style.css', LF will be replaced by CRLF the next time Git touches it

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (test)
$ git commit -m "1 commit"
[test 93dbb32] 1 commit
2 files changed, 3 insertions(+), 3 deletions(-)

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (test)
$ git checkout master
Switched to branch 'master'

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ vi Index.html

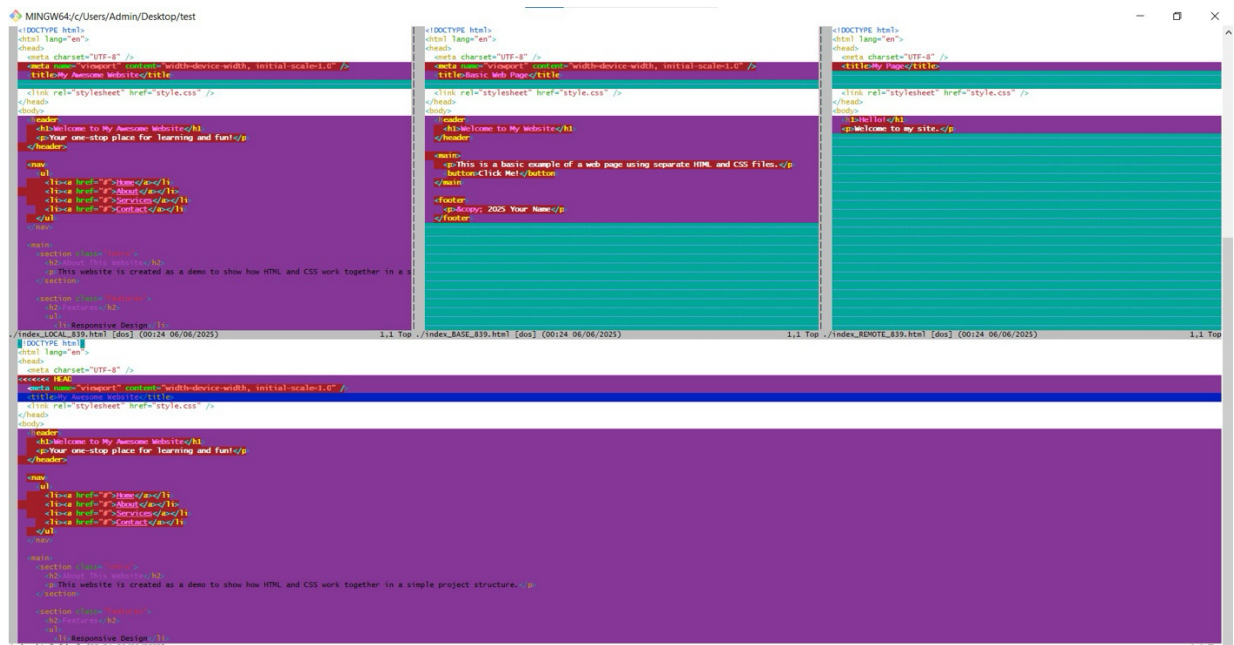
prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ vi Style.css

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ git add .

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ git commit -m "Master 2 commit"
[master bced1cf] Master 2 commit
2 files changed, 2 insertions(+), 2 deletions(-)

prave@LAPTOP-6KSGM7RP MINGW64 ~/Desktop/Project (master)
$ git merge test
Auto-merging Index.html
CONFLICT (content): Merge conflict in Index.html
Auto-merging Style.css
CONFLICT (content): Merge conflict in Style.css
Automatic merge failed; fix conflicts and then commit the result.

```

Handle a simple merge conflict

Normal merge conflict for 'Index.html':

{local}: modified file

{remote}: modified file

Hit return to start merge resolution tool (vimdiff):

4 files to edit

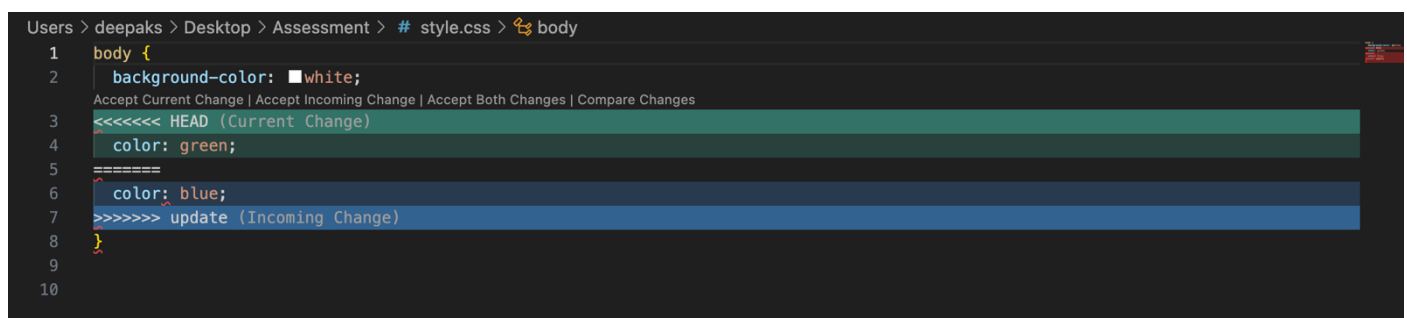
Normal merge conflict for 'style.css':

{local}: modified file

{remote}: modified file

Hit return to start merge resolution tool (vimdiff):

4 files to edit



```

1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>My Website</title>
5  </head>
6  <body>
   Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
7  <==== HEAD (Current Change)
8  <h1>Hello from main branch</h1>
9  =====
10 <h1>Hello from update branch</h1>
11 >>>>>> update (Incoming Change)
12 </body>
13 </html>
14
15 |

```

GIT IGNORE

```

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master|MERGING)
$ vi .env

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master|MERGING)
$ vi .log

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master|MERGING)
$ echo -e ".env\n".log >> .gitignore

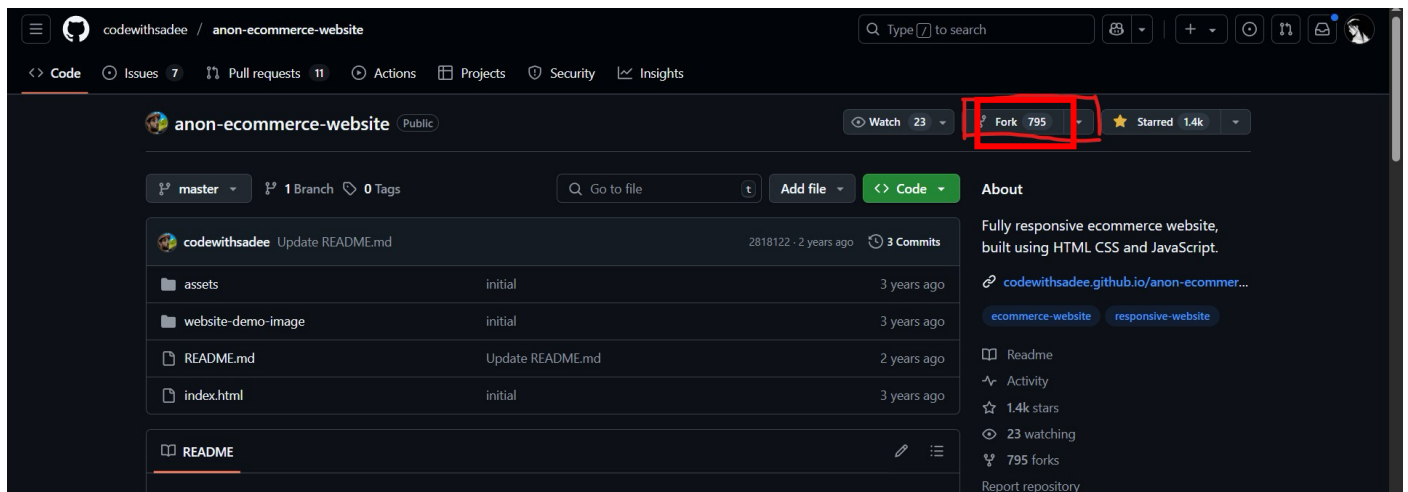
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master|MERGING)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master|MERGING)
$ git commit -m "Adding .gitignore"
[master 10f4e4d] Adding .gitignore

```

GIT CLONE

Navigate a project that you are interested in and fork it



The screenshot shows the GitHub interface for the repository 'anon-ecommerce-website' by user 'codewithsadee'. The repository is public and has 23 watchers, 795 forks, and 1.4k stars. The 'Fork' button is highlighted with a red box. The repository contains a 'master' branch with 1 branch and 0 tags. The file list includes 'assets', 'website-demo-image', 'README.md', and 'index.html'. The 'About' section describes it as a 'Fully responsive ecommerce website, built using HTML CSS and JavaScript' and provides a link to the repository on GitHub.

Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)

Required fields are marked with an asterisk (*).

Owner *

 Peeyush2005

Repository name *

/ anon-ecommerce-website

✅ anon-ecommerce-website is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description (optional)

Fully responsive ecommerce website, built using HTML CSS and JavaScript.

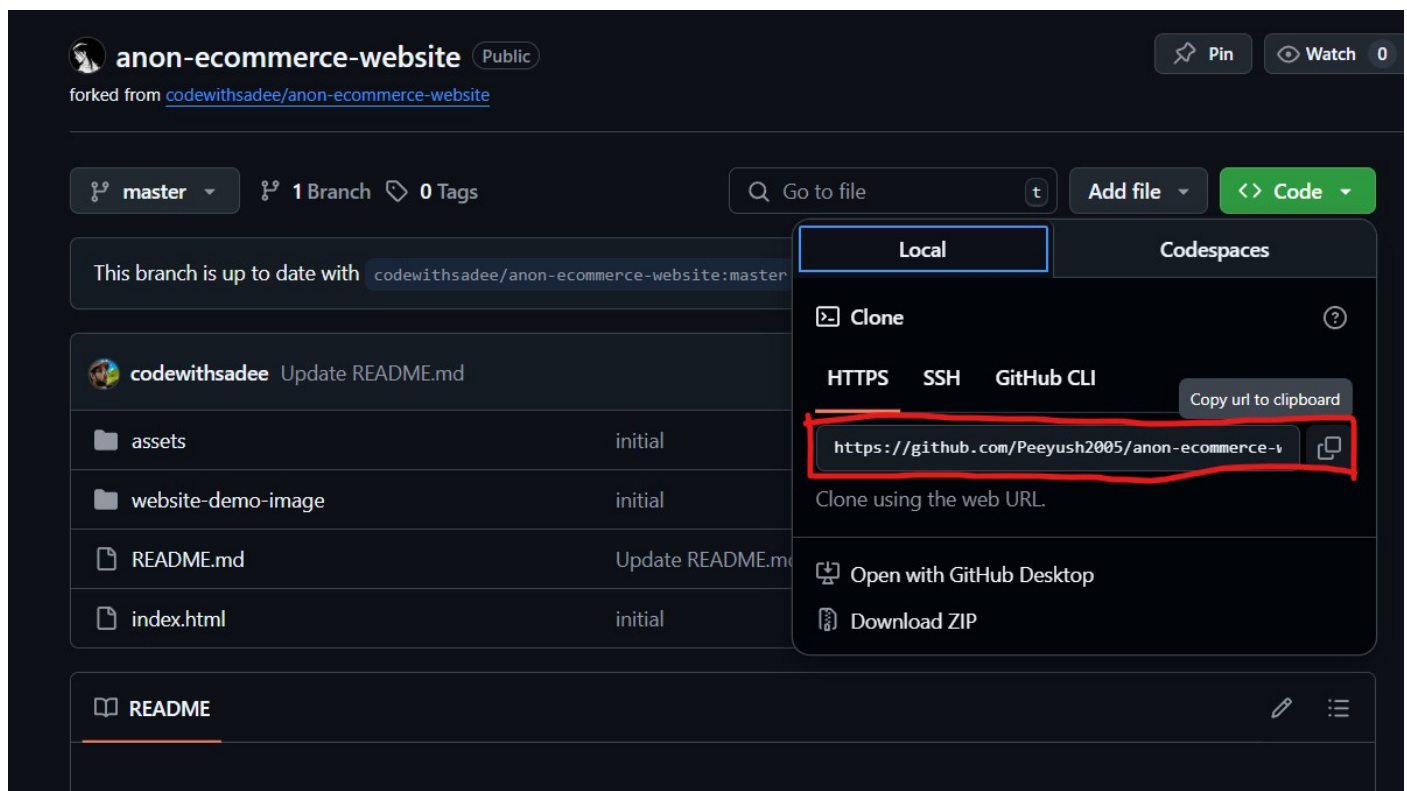
☐ Copy the `master` branch only

Contribute back to [codewithsadee/anon-ecommerce-website](#) by adding your own branch. [Learn more.](#)

📘 You are creating a fork in your personal account.

Create fork

Copy The URL



The screenshot shows the GitHub repository page for 'anon-ecommerce-website', which is a fork of 'codewithsadee/anon-ecommerce-website'. The repository is public and has 1 branch (master) and 0 tags. The 'Clone' button is highlighted, and the 'Local' tab is selected. The 'Clone' dropdown menu is open, showing the 'HTTPS' option selected. The URL 'https://github.com/Peeyush2005/anon-ecommerce-website' is displayed and highlighted with a red box. A 'Copy url to clipboard' button is also visible next to the URL. The repository files list shows 'assets', 'website-demo-image', 'README.md', and 'index.html'.

anon-ecommerce-website Public

forked from [codewithsadee/anon-ecommerce-website](#)

master 1 Branch 0 Tags

Go to file

Add file <> Code

This branch is up to date with [codewithsadee/anon-ecommerce-website:master](#)

 **codewithsadee** Update README.md

File	Commit
assets	initial
website-demo-image	initial
README.md	Update README.md
index.html	initial

README

Clone

HTTPS SSH GitHub CLI

Copy url to clipboard

https://github.com/Peeyush2005/anon-ecommerce-website

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

Go to git bash

```
Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ mkdir clone

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test (master)
$ cd clone

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test/clone (master)
$ git clone https://github.com/Peeyush2005/anon-ecommerce-website.git
Cloning into 'anon-ecommerce-website'...
remote: Enumerating objects: 97, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 97 (delta 0), reused 0 (delta 0), pack-reused 93 (from 1)
Receiving objects: 100% (97/97), 4.42 MiB | 5.17 MiB/s, done.
Resolving deltas: 100% (8/8), done.

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test/clone (master)
$ cd anon-ecommerce-website

Admin@DESKTOP-OD400I1 MINGW64 ~/Desktop/test/clone/anon-ecommerce-website (master)
$ ls
README.md  assets/  index.html  website-demo-image/
```

Now the file can be accessed